

304 SS RIS and Oxidation Prediction by Off-Lattice KMC Model

Zhuo Wang

March 24, 2026

1 Introduction

2 Simulation Method

In this study, an off-lattice kinetic Monte Carlo (olKMC) model is employed to simulate radiation-induced segregation (RIS) and grain boundary oxidation in 304 stainless steel (304 SS). The simulation framework consists of two main parts: elemental diffusion and grain boundary oxidation. In the following subsections, the modeling approach for RIS and the treatment of oxidation processes are described separately.

2.1 Radiation Induced Segregation

To phenomenologically capture RIS, we employ an effective chemical potential that accounts for the preferential enrichment or depletion of alloying elements to grain boundaries under irradiation. The diffusion flux of alloying species driven by the chemical potential gradient is expressed as

$$J_i = -\frac{D_i C_i}{k_B T} \frac{\partial (\mu_0 + \Delta\mu_i(x))}{\partial x} \quad (1)$$

where J_i is the diffusion flux of species i , D_i is the diffusion coefficient, C_i is the concentration, k_B is the Boltzmann constant, T is the absolute temperature, μ_0 is the reference chemical potential, and $\Delta\mu_i(x)$ is the radiation-induced effective chemical potential contribution.

The fluxes satisfy the mass-balance constraint

$$\sum_i J_i + J_V = 0 \quad (2)$$

where J_V is the vacancy flux, given by Fick's first law:

$$J_V = -D_V \frac{\partial C_V}{\partial x} \quad (3)$$

with D_V and C_V denoting the vacancy diffusion coefficient and vacancy concentration, respectively.

For irradiated 304 stainless steel (304 SS), vacancies are continuously generated in the grain interior, whereas the grain boundary serves as a strong vacancy sink. Accordingly, the vacancy concentration profile near the grain boundary is described by

$$C_V(x) = C_V^\infty(\text{dose}) - (C_V^\infty(\text{dose}) - C_V^{\text{GB}}) \exp(-kx) \quad (4)$$

where x is the distance from the grain boundary, $C_V^\infty(\text{dose})$ is the bulk vacancy concentration at a given irradiation dose, C_V^{GB} is the vacancy concentration at the grain boundary, and k is the decay constant. In the present work, $C_V^{\text{GB}} = 0$ is assumed. Thus the vacancy flux is within the form

$$J_V = -kD_VC_V^\infty(\text{dose}) \exp(-kx) \quad (5)$$

The mass-balance constraint (Eq. 2) can be rewritten as:

$$\sum_i J_i = kD_VC_V^\infty(\text{dose}) \exp(-kx) \quad (6)$$

By intergrating the two sides of 6, the mass-balance constraint for RIS effective chemical potential can be expressed as:

$$\sum_i D_i C_i \Delta \mu_i(x) = K_B T D_V C_V^\infty(\text{dose}) \exp(-kx) + \text{Const} \quad (7)$$

We assume that the effective chemical potential is zero at infinite distance from the grain boundary, thus the constant term in Eq. 7 is zero. The effective chemical potential constraint by mass balance can be expressed as :

$$\sum_i D_i C_i \Delta \mu_i(x) = K_B T D_V C_V^\infty(\text{dose}) \exp(-kx) \quad (8)$$

The effective chemical potential contribution for each alloying element is assumed to follow an exponential decay form:

$$\Delta\mu_i(x) = \Delta\mu_i^{\text{GB}} \exp\left(-\frac{x}{\lambda}\right) \quad (9)$$

where $\Delta\mu_i^{\text{GB}}$ is the effective chemical potential contribution at the grain boundary, and λ is the characteristic length scale. For 304 SS, the segregation of Ni and Cr are mainly contributed by Inverse Kirkendall effect, $\Delta\mu_i^{\text{GB}}$ is given by:

$$\Delta\mu_i^{\text{GB}} = \Delta\mu_0^{\text{GB}} \ln \frac{D_i}{D_{Fe}} \quad (10)$$

where D_{Fe} is the diffusion coefficient of Fe, and $\Delta\mu_0^{\text{GB}}$ is a fitting parameter that represents the overall magnitude of the effective chemical potential contribution at the grain boundary. $D_{Cr} > D_{Fe} > D_{Ni}$, therefore $\Delta\mu_{Cr}^{\text{GB}} > 0$ and $\Delta\mu_{Ni}^{\text{GB}} < 0$, which captures the enrichment of Cr and depletion of Ni at the grain boundary under irradiation. For Si, the segregation is mainly contributed by forming solute-interstitial complexes. The effective chemical potential contribution for Si is given by:

$$\Delta\mu_{Si}^{\text{GB}} = \Delta\mu_0^{\text{GB}} \ln \frac{C_{SiI}}{C_{Si}} \quad (11)$$

where C_{SiI} is the concentration of Si-interstitial complexes, which is assumed to be proportional to the interstitial concentration C_I and the Si concentration C_{Si} :

$$C_{SiI} = KC_IC_{Si} \quad (12)$$

where K is the equilibrium constant for the formation of Si-interstitial complexes. The interstitial concentration profile is assumed to follow a similar form as the vacancy concentration profile:

$$C_I(x) = C_I^\infty(\text{dose}) - (C_I^\infty(\text{dose}) - C_I^{\text{GB}}) \exp(-k'x) \quad (13)$$

where $C_I^\infty(\text{dose})$ is the bulk interstitial concentration at a given irradiation dose, C_I^{GB} is the interstitial concentration at the grain boundary, and k' is the decay constant. In the present work, $C_I^{\text{GB}} = 0$ is assumed. Thus the interstitial concentration profile can be expressed as:

$$C_I(x) = C_I^\infty(\text{dose}) (1 - \exp(-k'x)) \quad (14)$$

By substituting Eq. 14 into Eq. 11, the effective chemical potential contribution for Si can be expressed as: